*CASE1:*Electricity Bill

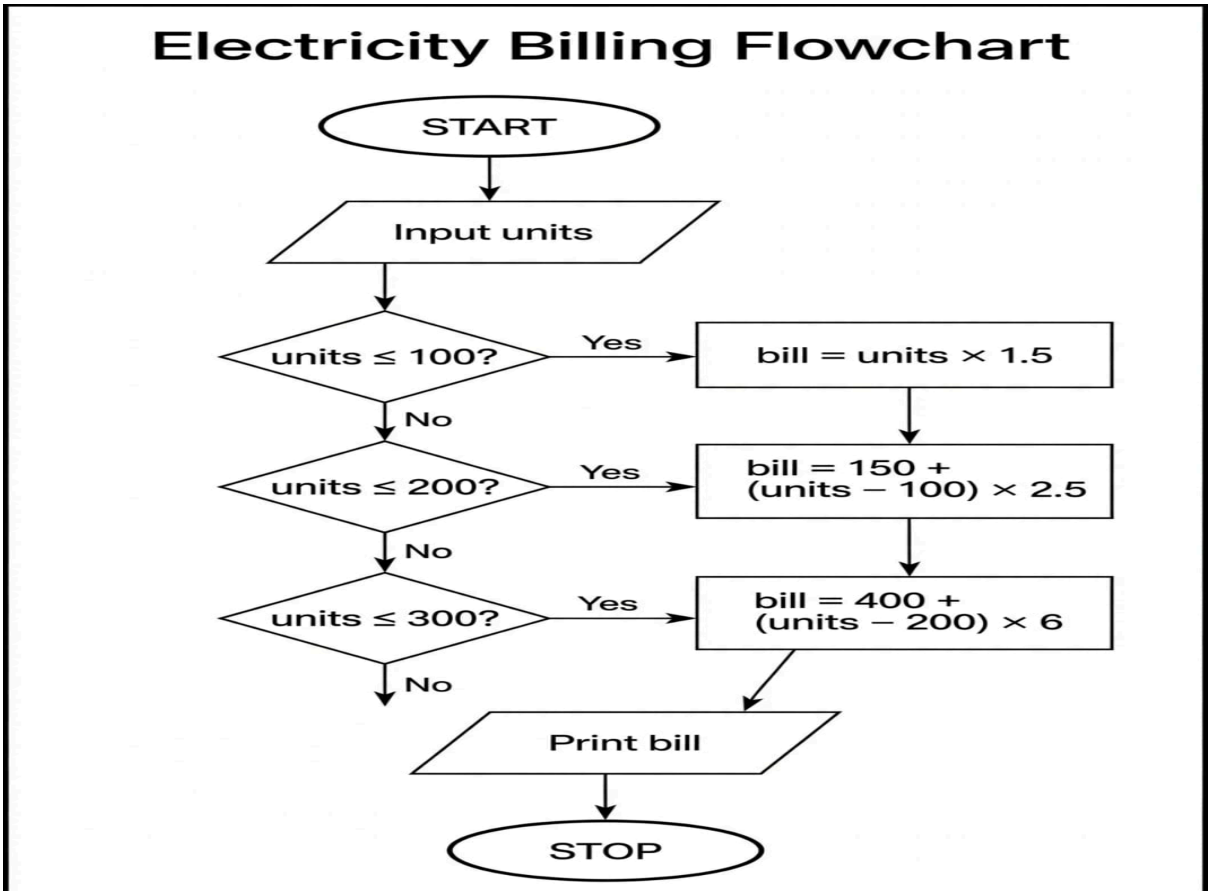
Problem Statement: Calculate electricity bill

Objective:We use if-elif condition in this program.

Algorithm for Electricity bill:

1.Start the program
2.Ask user for units
3.Check which slab the units fall
4.Calculate bill as per slab rate
5.Show bill amount
6.Stop

Flowchart:



Code:

```
units=int(input("Enter units: "))
if units<=100:
   bill=units*1.5
elif units<=200:
    bill =100*1.5+(units-100)*2.5
elif units<=300:
    bill=100*1.5+100*2.5+(units-200)*4
else:
   bill=100*1.5+100*2.5+100*4+(units-300)*6
print(f"Bill :Rs.{bill}")
```

CASE1:Electricity Bill

Code:

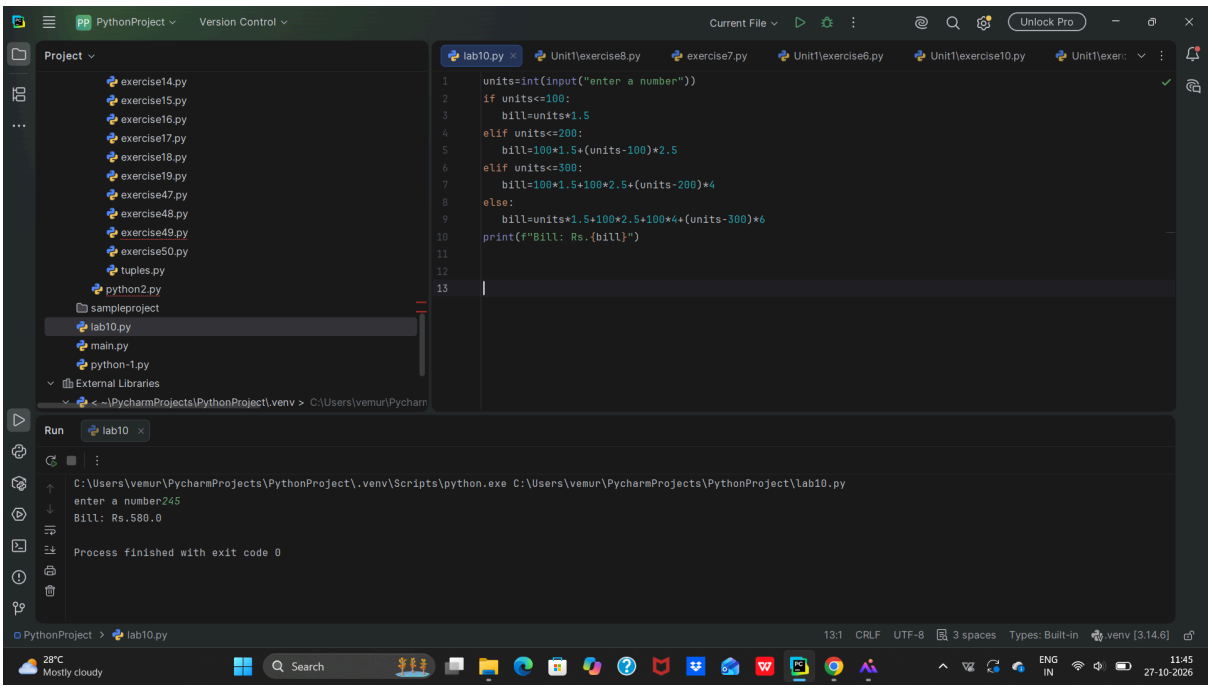
```
units=int(input("enter a number"))
if units<=100:
  bill=units*1.5
elif units<=200:
  bill=100*1.5+(units-100)*2.5
elif units<=300:
  bill=100*1.5+100*2.5+(units-200)*4
else:
  bill=units*1.5+100*2.5+100*4+(units-300)*6
print(f"Bill: Rs.{bill}")
```

Output:

Enter units: 234

Bill :Rs.536.0

Screenshot:



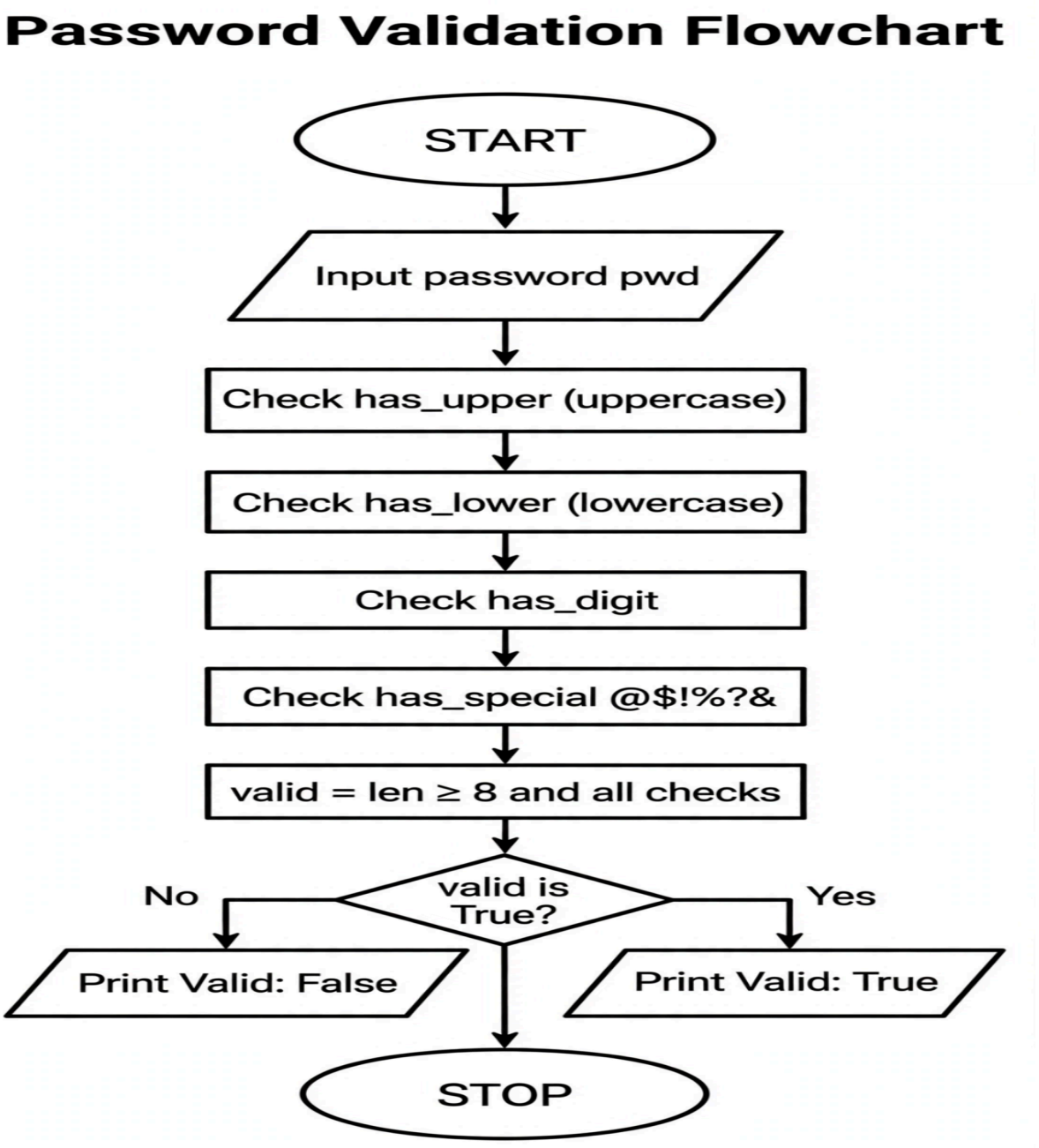
CASE2:Password Validation

Problem Statement: Calculate password validation

Algorithm for password validation:

1.Start
2.Take password from users
3.Check password has capital letters,small letters,numbers and symbols
4.Check password length is 8 or more
5.If all true - Valid
6.ELSE - Not Valid
7.Print result
8.Stop

Flowchart:



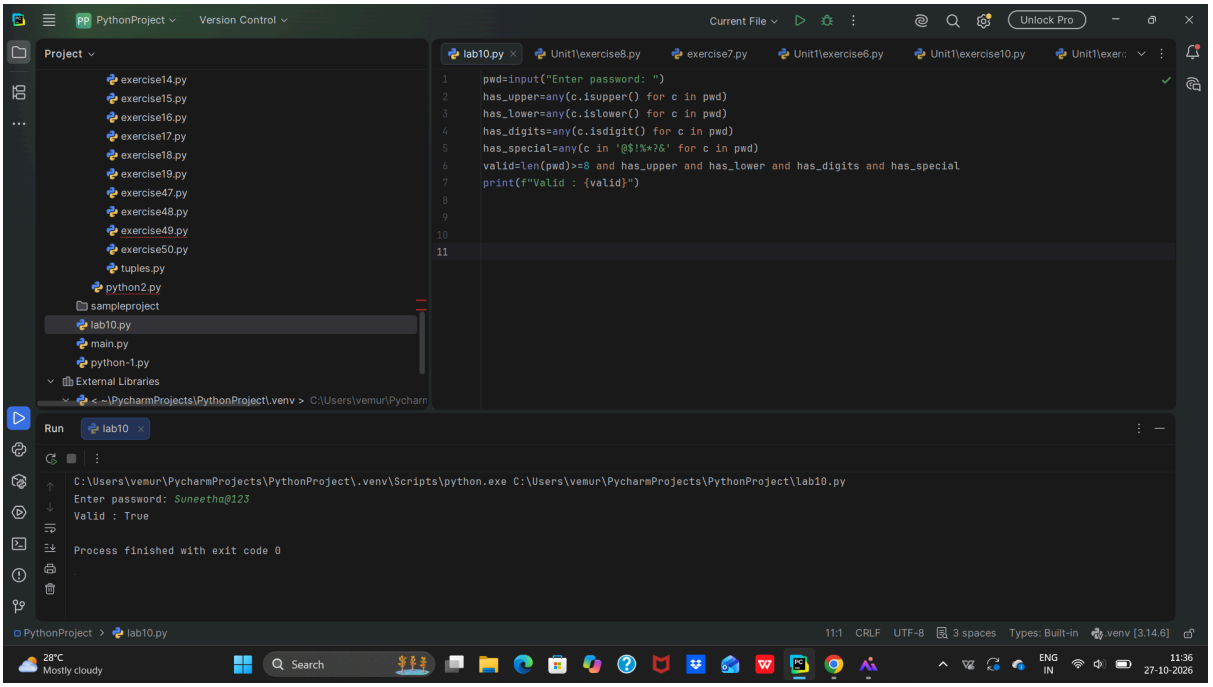
CASE2:Password Validation

Code:

```
pwd=input("Enter password: ")
has_upper=any(c.isupper() for c in pwd)
has_lower=any(c.islower() for c in pwd)
has_digits=any(c.isdigit() for c in pwd)
has_special=any(c in '@$!%*?&' for c in pwd)
valid=len(pwd)>=8 and has_upper and has_lower and has_digits and has_special
print(f"Valid : {valid}")
```

Output:
Enter password: Suneetha@123
Valid : True

Screenshot:



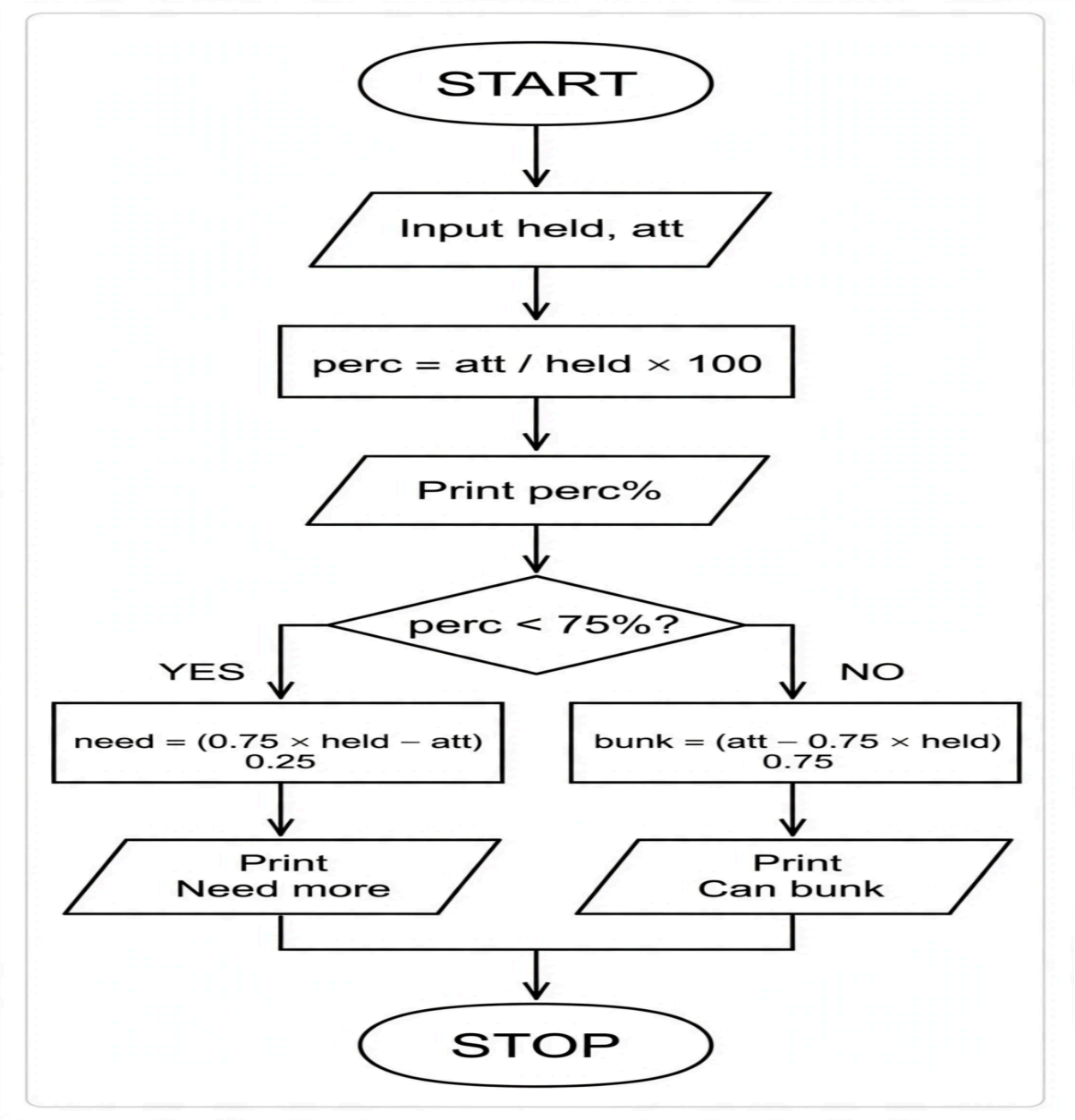
CASE3:Attendance Recovery

Problem Statement: Calculate Attendance recovery

Algorithm for attendance recovery:

1.Start
2.Enter held and attended
3.Find % = att/held*100
4.If % <75 show need,else show bunk
5.Stop

Flowchart:



CASE3:Attendance Recovery

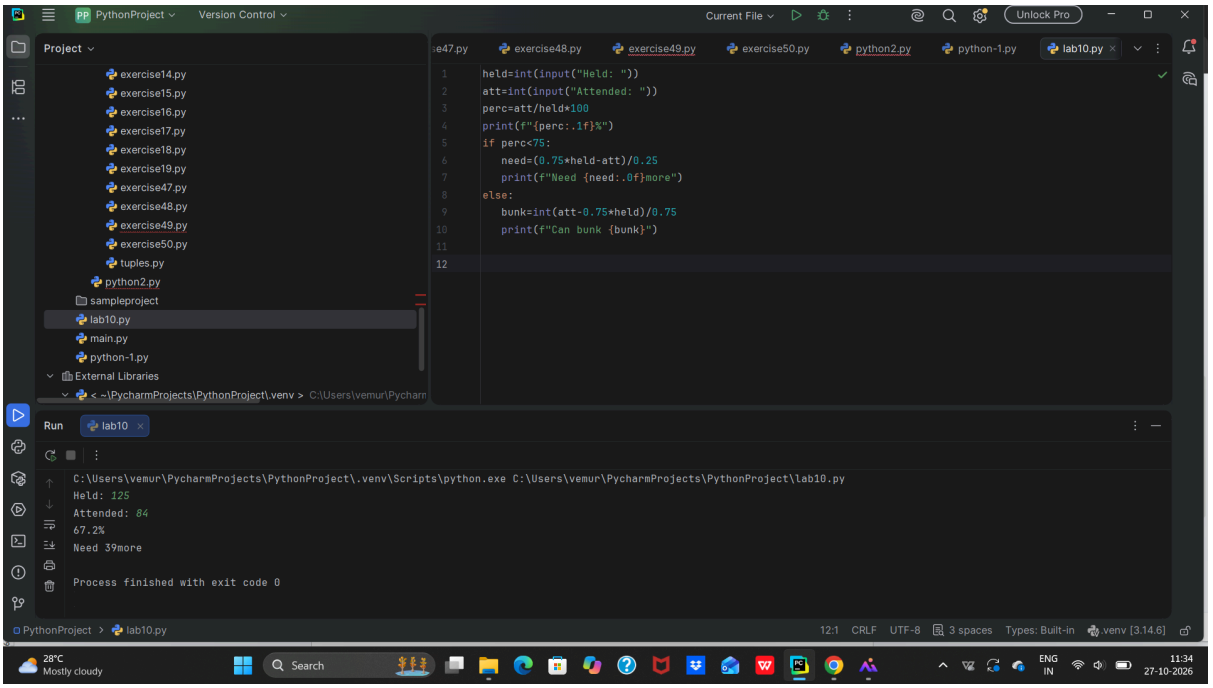
Code:

```
held=int(input("Held: "))
att=int(input("Attended: "))
perc=att/held*100
print(f"{perc:.1f}%")
if perc<75:
   need=(0.75*held-att)/0.25
   print(f"Need {need:.0f}more")
else:
   bunk=int(att-0.75*held)/0.75
   print(f"Can bunk {bunk}")
```

Output:

Held: 125
Attended: 84
67.2%
Need 39more

Screenshot:



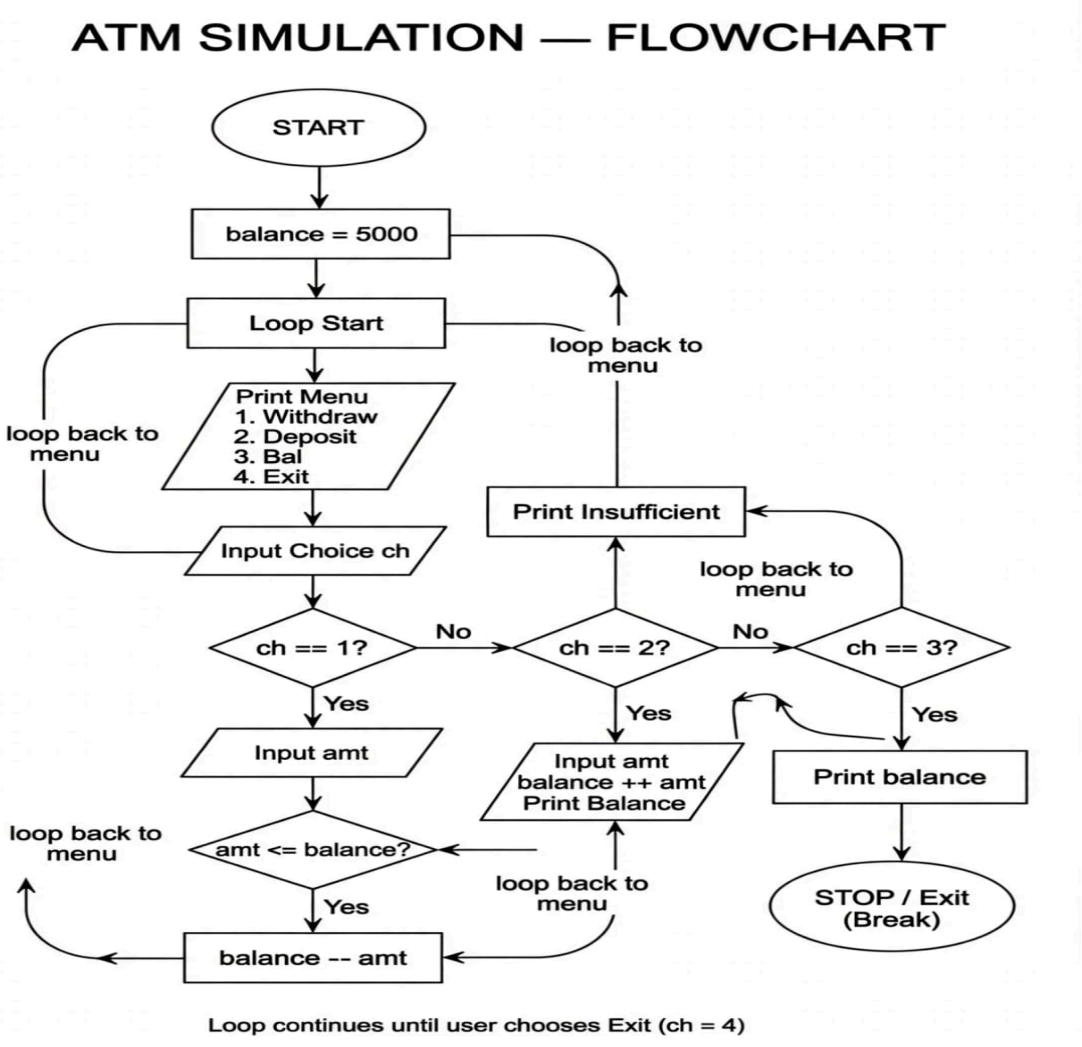
CASE4:ATM Simulation

Problem Statement: Calculate ATM Simulation

Algorithm for atm simulation:

1.Start
2.balance=5000
3.Print menu and take choice
4.If choice=1 check balance and withdraw
5.If choice=2 add amount to balance
6.If choice=3 print balance
7.If choice=4 exit loop
8.Stop

Flowchart:



CASE4:ATM Simulation

Code:

```
balance=5000
while True:
   print("1.Withdraw 2.Deposit 3.Bal 4.Exit")
   ch=int(input("Choice: "))
   if ch==1:
       amt=int(input("Amount: "))
       if amt<=balance:
           balance=balance-amt
           print(f"New Balance {balance}")
           print()
       else:
           print("Insufficient Balance")
           print()
   elif ch==2:
       amt=int(input("Amount: "))
       balance=balance+amt
       print(f"Balance {balance}")
       print()
   elif ch==3:
        print(f"Balance {balance}")
        print()
   else:
       print("Exit")
       break
```

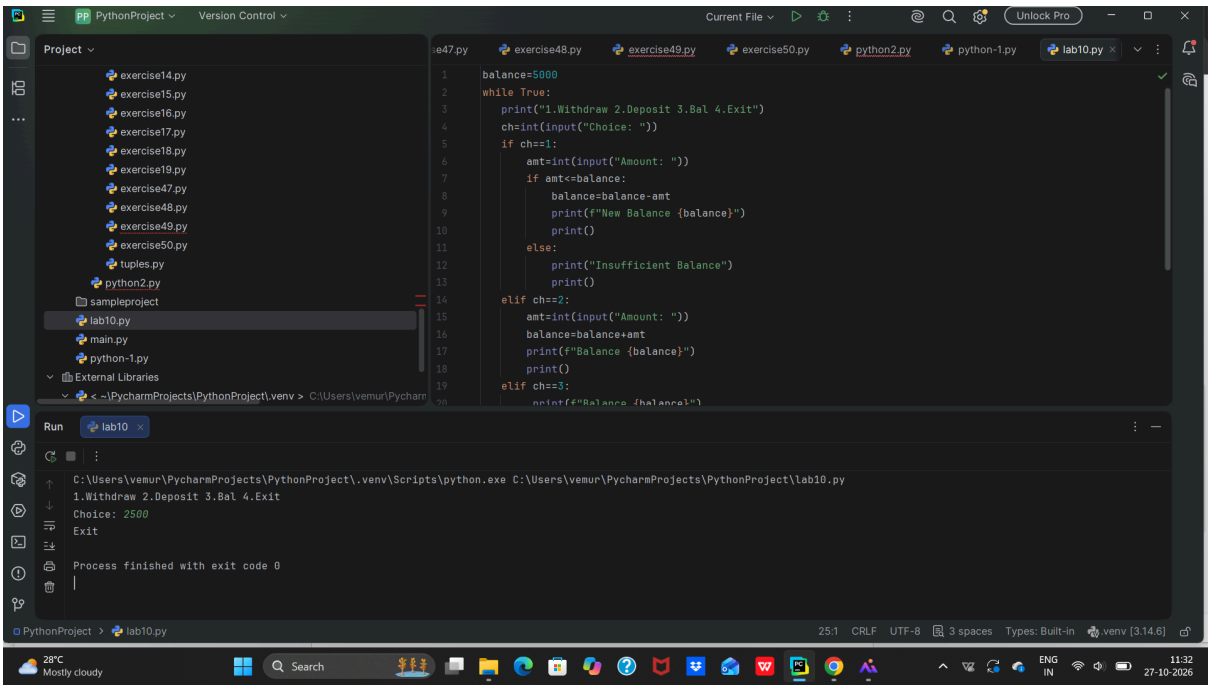
Output:
1.Withdraw 2.Deposit 3.Bal 4.Exit
Choice: 1
Amount: 1000
New Balance 4000

1.Withdraw 2.Deposit 3.Bal 4.Exit
Choice: 2
Amount: 500
Balance 4500

1.Withdraw 2.Deposit 3.Bal 4.Exit
Choice: 4
Exit

Screenshot:

CASE4:ATM Simulation



CASE5:Student Grading

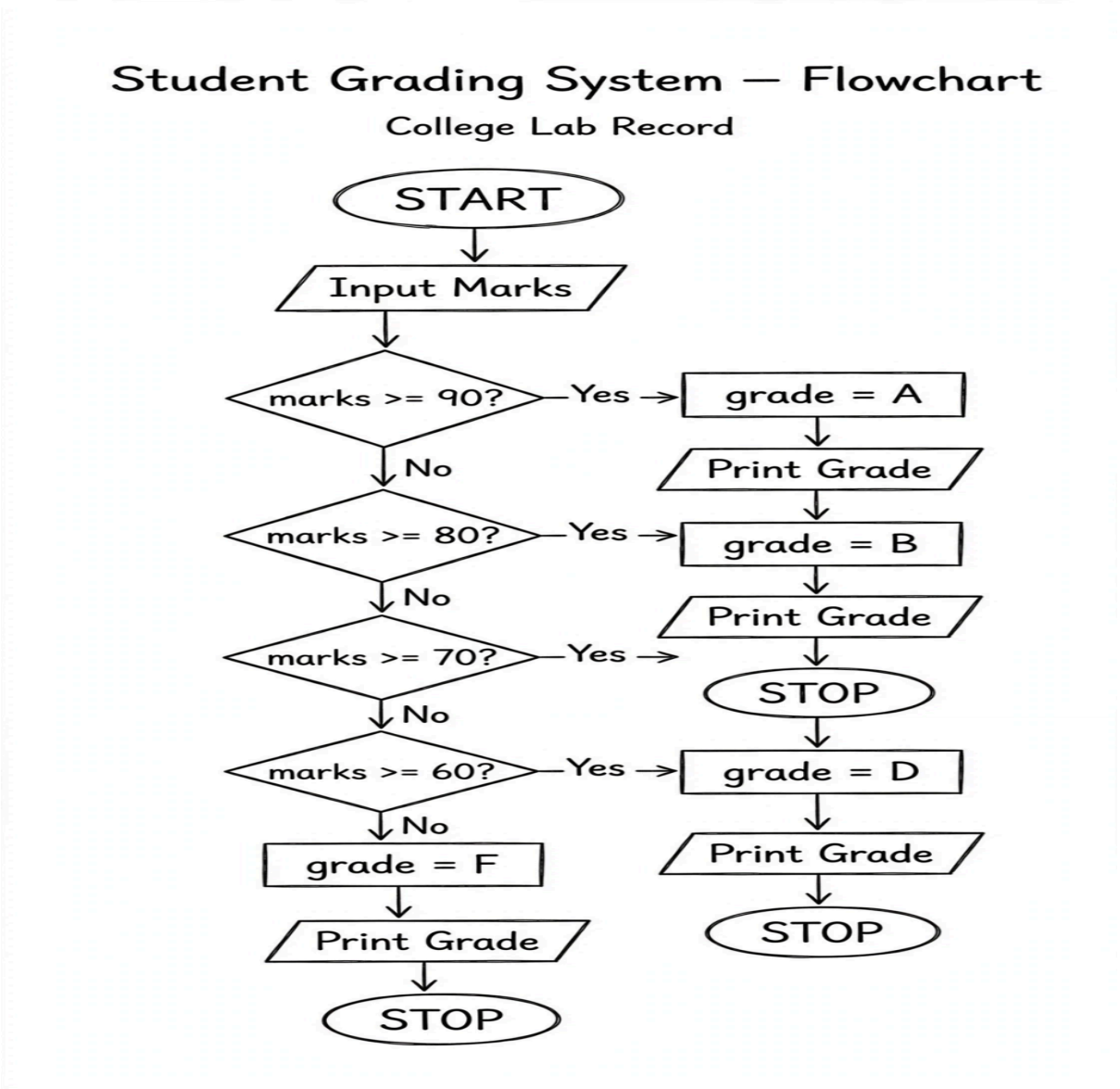
Problem Statement: Calculate student grading

Objective:We use if-elif condition in this program.

Algorithm for student grading:

1.Start the program
2.Input marks
3.Check marks by using if-elif ladder
4.Print grade
5.Stop

Flowchart:



CASE5:Student Grading

Code:

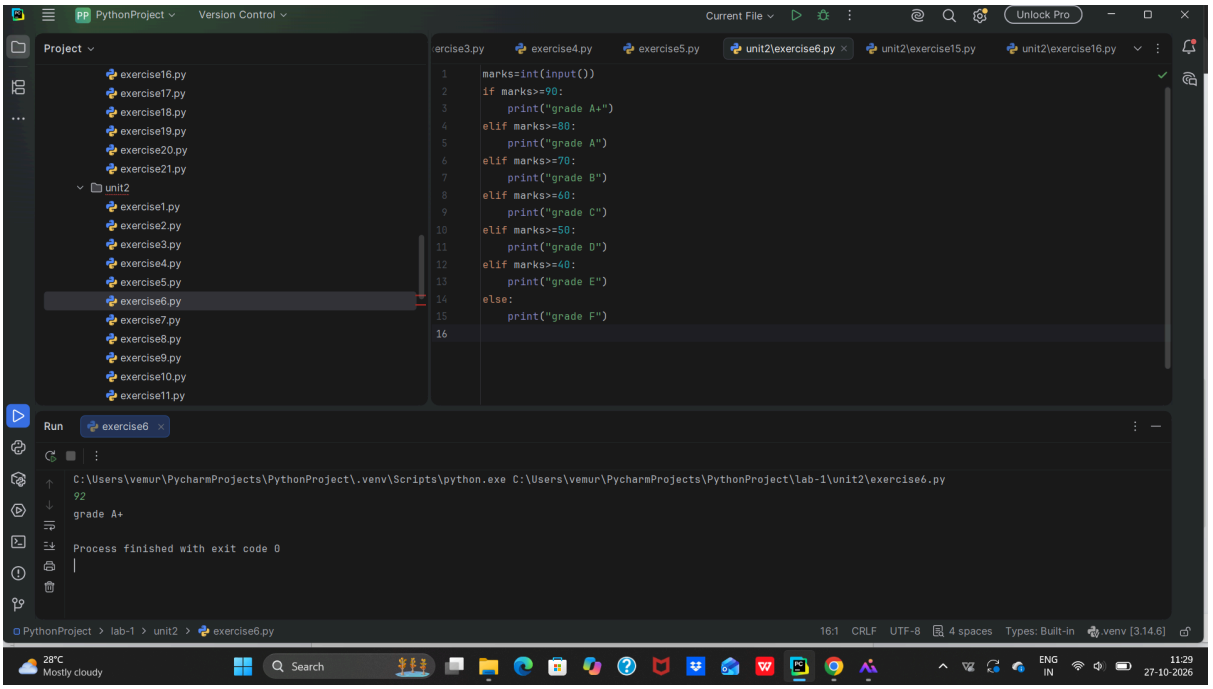
```
marks=int(input())
if marks>=90:
  grade="A"
elif marks>=80:
  grade="B"
elif marks>=70:
   grade="C"
elif marks>=60:
   grade="D"
else:
   grade="F"
print(f"Grade:{grade}")
```

Output:

85

Grade:B

Screenshot:



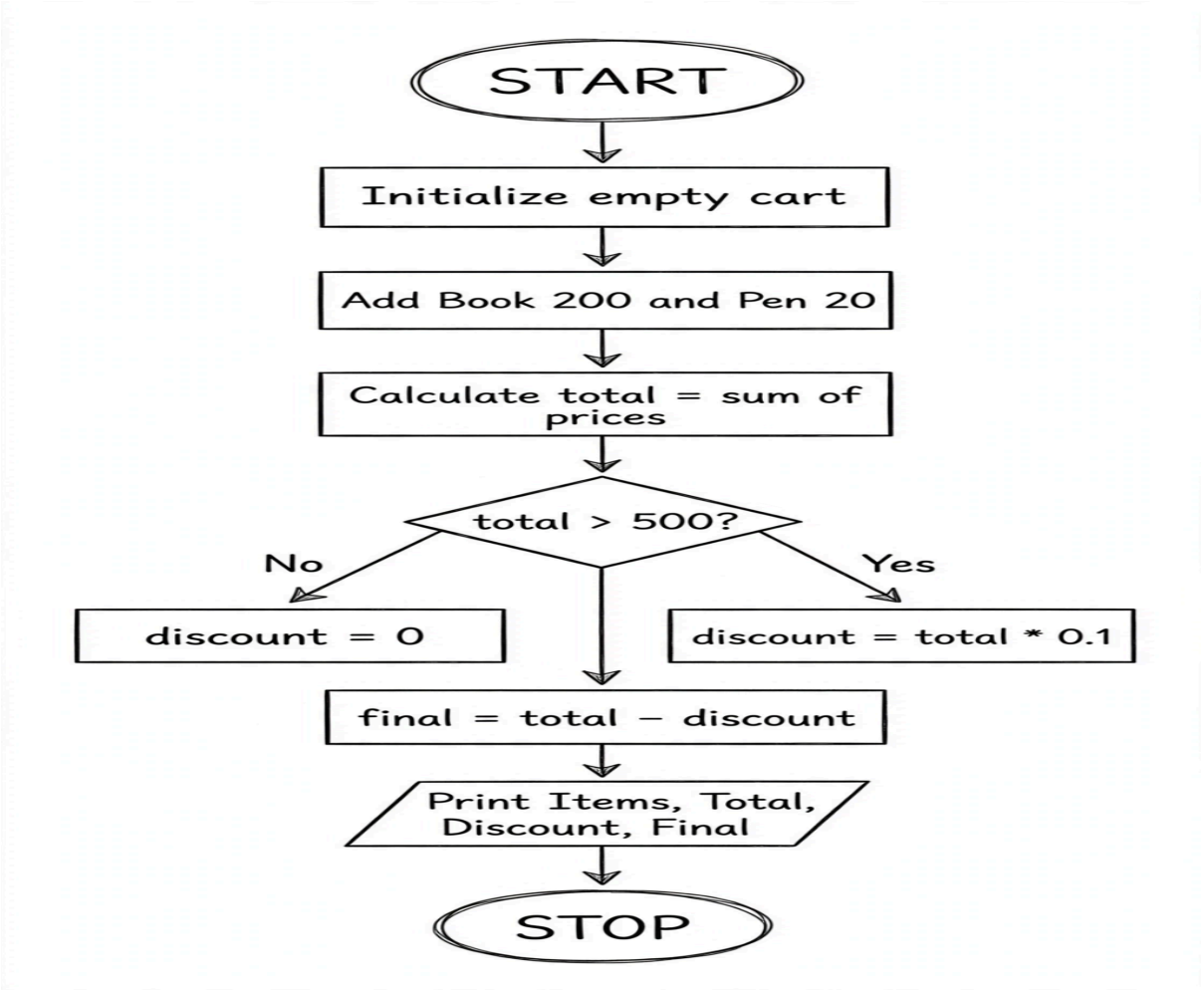
CASE6:Shopping Cart

Problem Statement: Calculate shopping cart

Algorithm for shopping cart:

1.Start
2.Create empty cart[ ]
3.Add Books=200 and Pen=20 to cart
4.Total =sum of all prices
5.If total>500 THEN discount=total*0.1 ELSE discount=0
6.final = total-discount
7.Print cart,total,discount,final
8.Stop

Flowchart:



Code:

```
cart=[]
cart.append({"item":"Book","price" :200})
cart.append({"item":"Pen","price":20})
```

CASE6:Shopping Cart

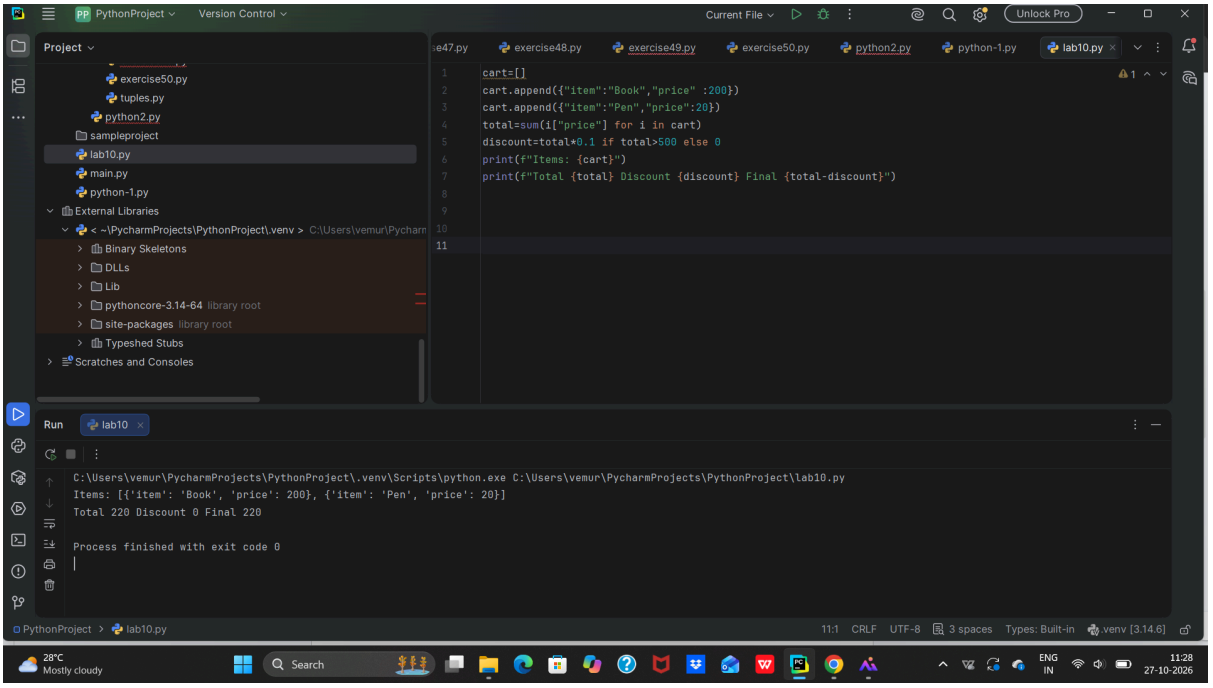
```
total=sum(i["price"] for i in cart)
discount=total*0.1 if total>500 else 0
print(f"Items: {cart}")
print(f"Total {total} Discount {discount} Final {total-discount}")
```

Output:

Items: [{'item': 'Book', 'price': 200}, {'item': 'Pen', 'price': 20}]

Total 220 Discount 0 Final 220

Screenshot:



CASE7:Leap Year

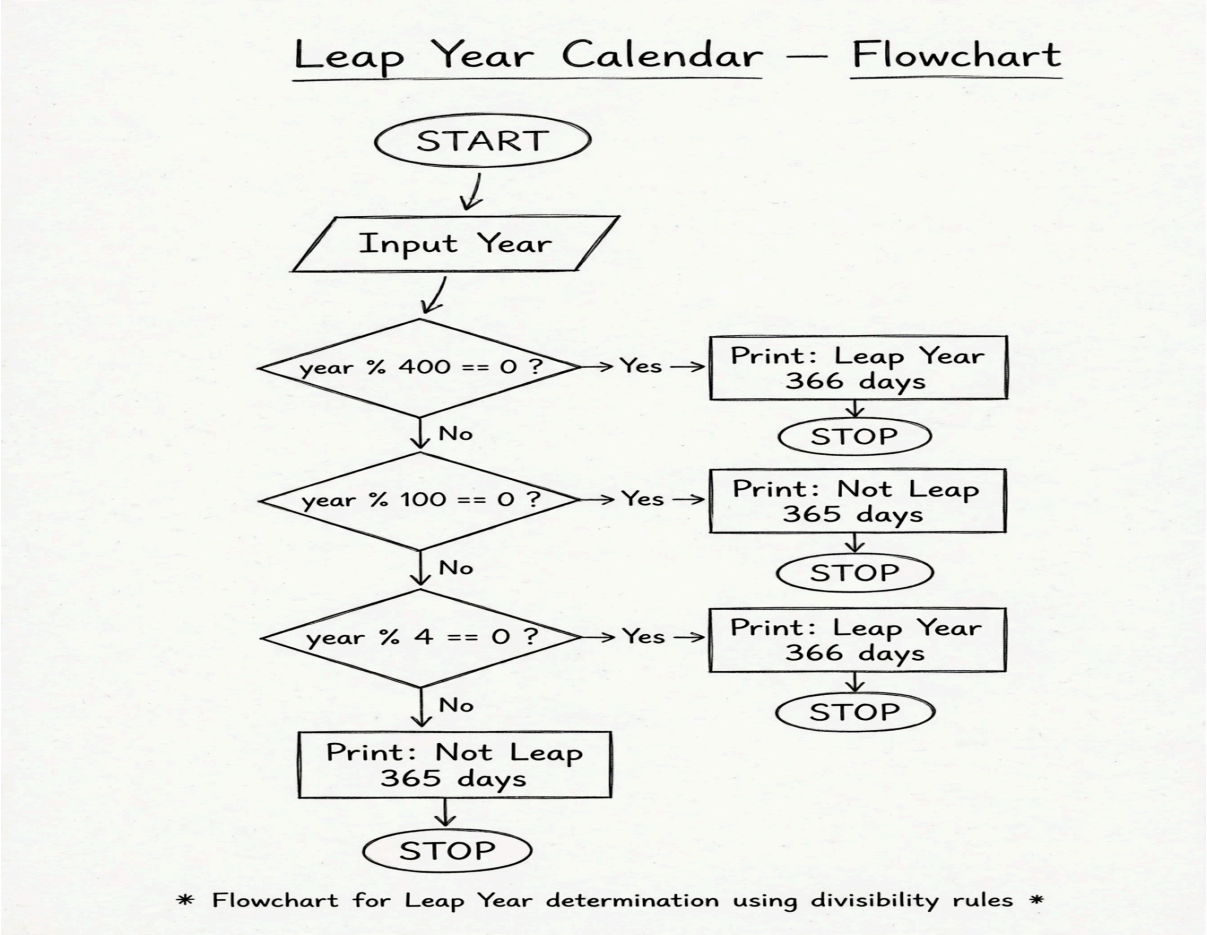
Problem Statement: Calculate leap year

Objective:We use if condition in this program.

Algorithm for leap year:

1.Start the program
2.Read year
3.Check if year %400==0
yes->Leap year
No->Check if 100==0->Not leap
No->Check if year %4==0->Leap else Not Leap
4.Print result
5.Stop

Flowchart:



Code:

```
year = int(input("Year: "))
if year%4== 0 and year % 100 != 0 or year % 400 == 0:
   print(f"{year} is leap year-366 days")
else:
  print(f"{year} is not a leap year-365 days")
```

Output:

Year: 2024
2024 is leap year-366 days

CASE7:Leap Year

Code:

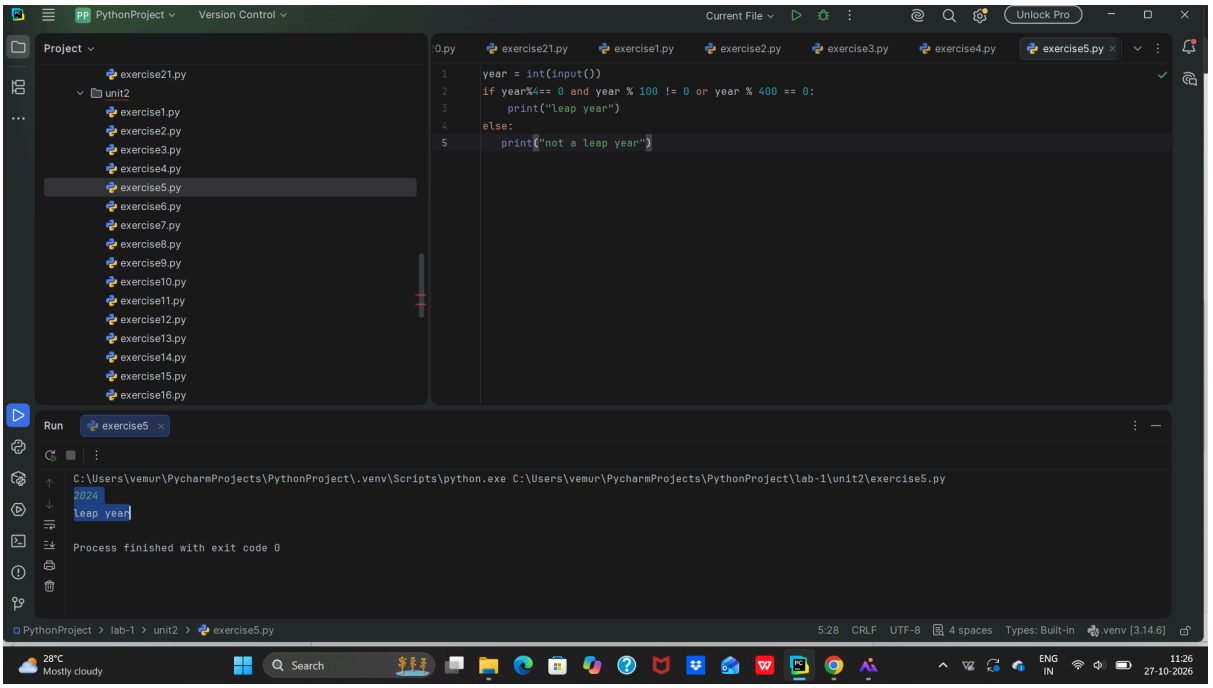
```
year = int(input())
if year%4== 0 and year % 100 != 0 or year % 400 == 0:
   print("leap year")
else:
  print("not a leap year")
```

Output:

2024

leap year

Screenshot:



Code:

```
year = int(input("Year: "))
if year%4== 0 and year % 100 != 0 or year % 400 == 0:
   print(f"{year} is leap year-366 days")
else:
  print(f"{year} is not a leap year-365 days")
```

Output:

Year: 2024

2024 is leap year-366 days

CASE8:Basic Calculator

Problem Statement: Calculate Basic calculator

Objective:We use if-elif condition in this.

Algorithm for Basic Calculator:
:

1.Start the program
2.Ask user for units
3.Check which slab the units fall
4.Calculate bill as per slab rate
5.Show bill amount
6.Stop

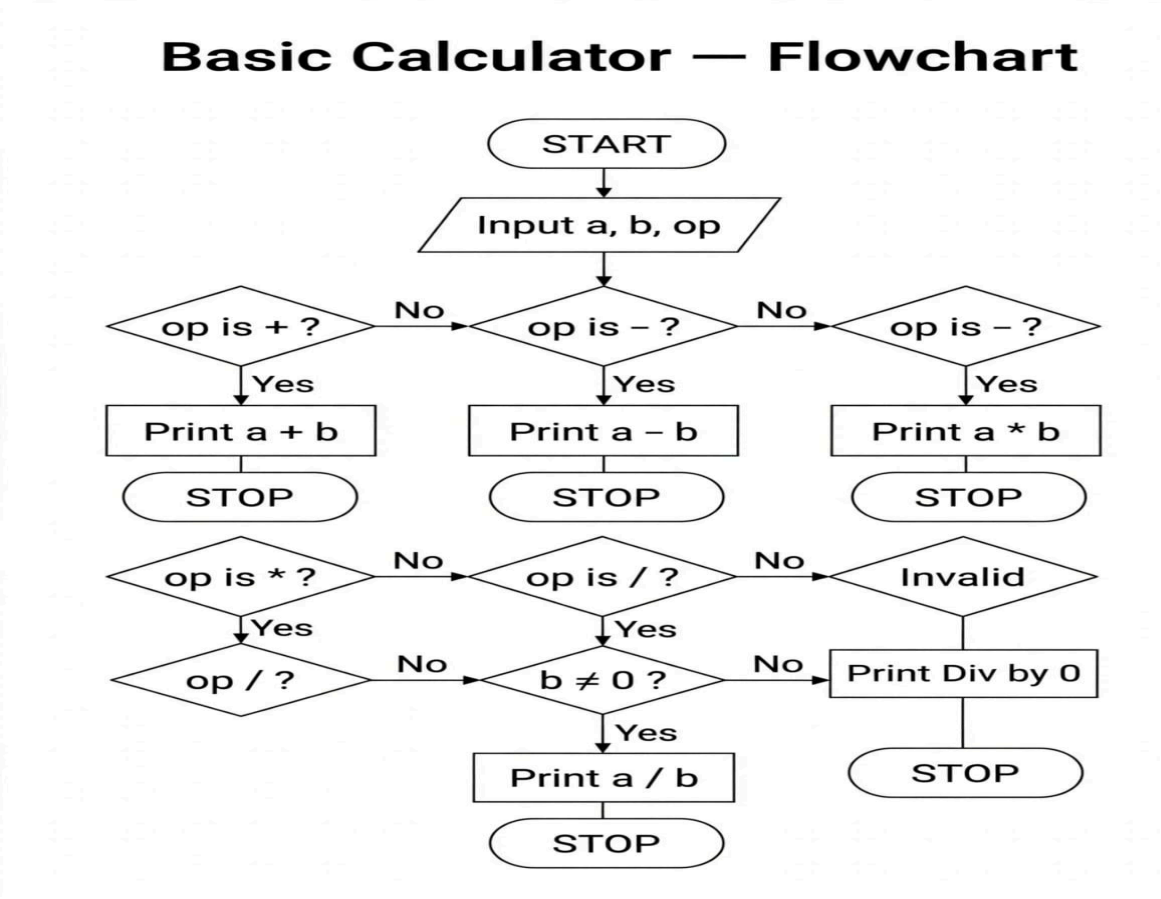
Flowchart:

Code:

```
a=int(input("a: "))
b=int(input("b: "))
op=input("Op + - * / : ")
if op=="+":
   print(a+b)
elif op=="-":
   print(a-b)
elif op=="*":
   print(a*b)
elif op=="/":
   print(a/b if b!=0 else "Div by 0")
```

Output:
a: 10
b: 5
Op + - * / : +
15

CASE8:Basic Calculator



Code:

```
a=int(input("a: "))
b=int(input("b: "))
op=input("Op + - * / : ")
if op=="+":
  print(a+b)
elif op=="-":
```

Code:

```
a=int(input("a: "))
b=int(input("b: "))
op=input("Op + - * / : ")
if op=="+":
   print(a+b)
elif op=="-":
   print(a-b)
elif op=="*":
   print(a*b)
elif op=="/":
   print(a/b if b!=0 else "Div by 0")
```

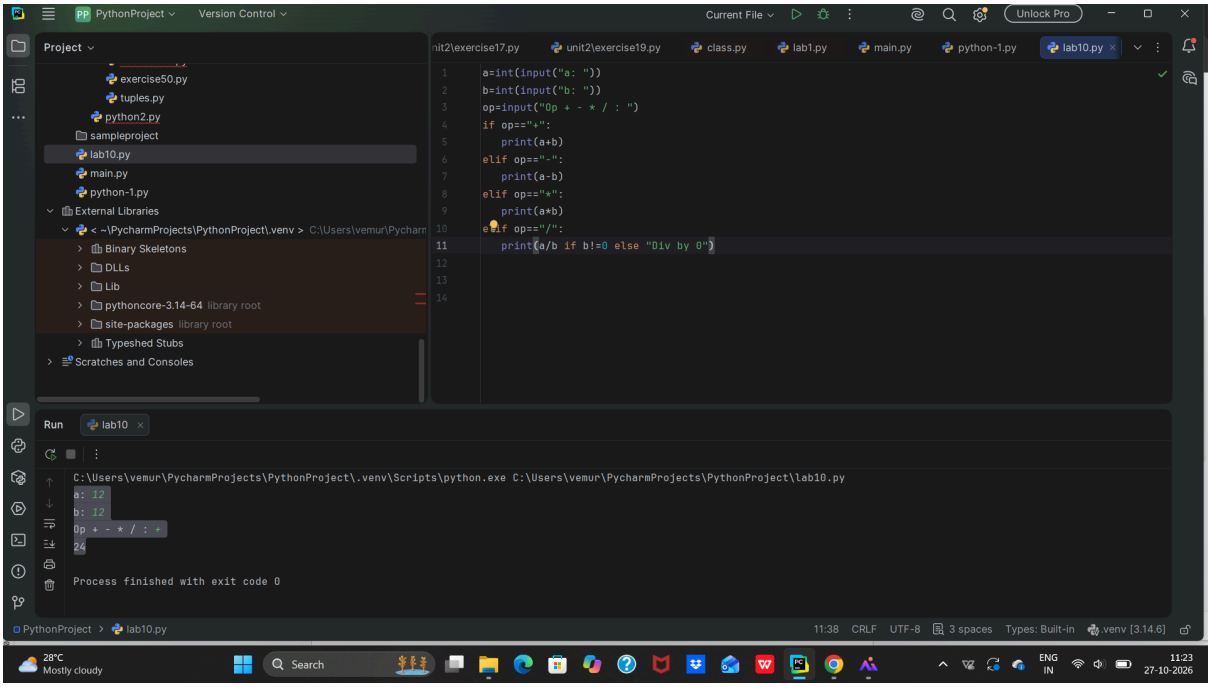
Output:
a: 10
b: 5
Op + - * / : +
15

CASE8:Basic Calculator

```
  print(a-b)
elif op=="*":
  print(a*b)
elif op=="/":
  print(a/b if b!=0 else "Div by 0")
```

Output:
a: 12
b: 12
Op + - * / : +
24

Screenshot:



Code:

```
a=int(input("a: "))
b=int(input("b: "))
op=input("Op + - * / : ")
if op=="+":
   print(a+b)
elif op=="-":
   print(a-b)
elif op=="*":
   print(a*b)
elif op=="/":
   print(a/b if b!=0 else "Div by 0")
```

Output:
a: 10
b: 5
Op + - * / : +
15

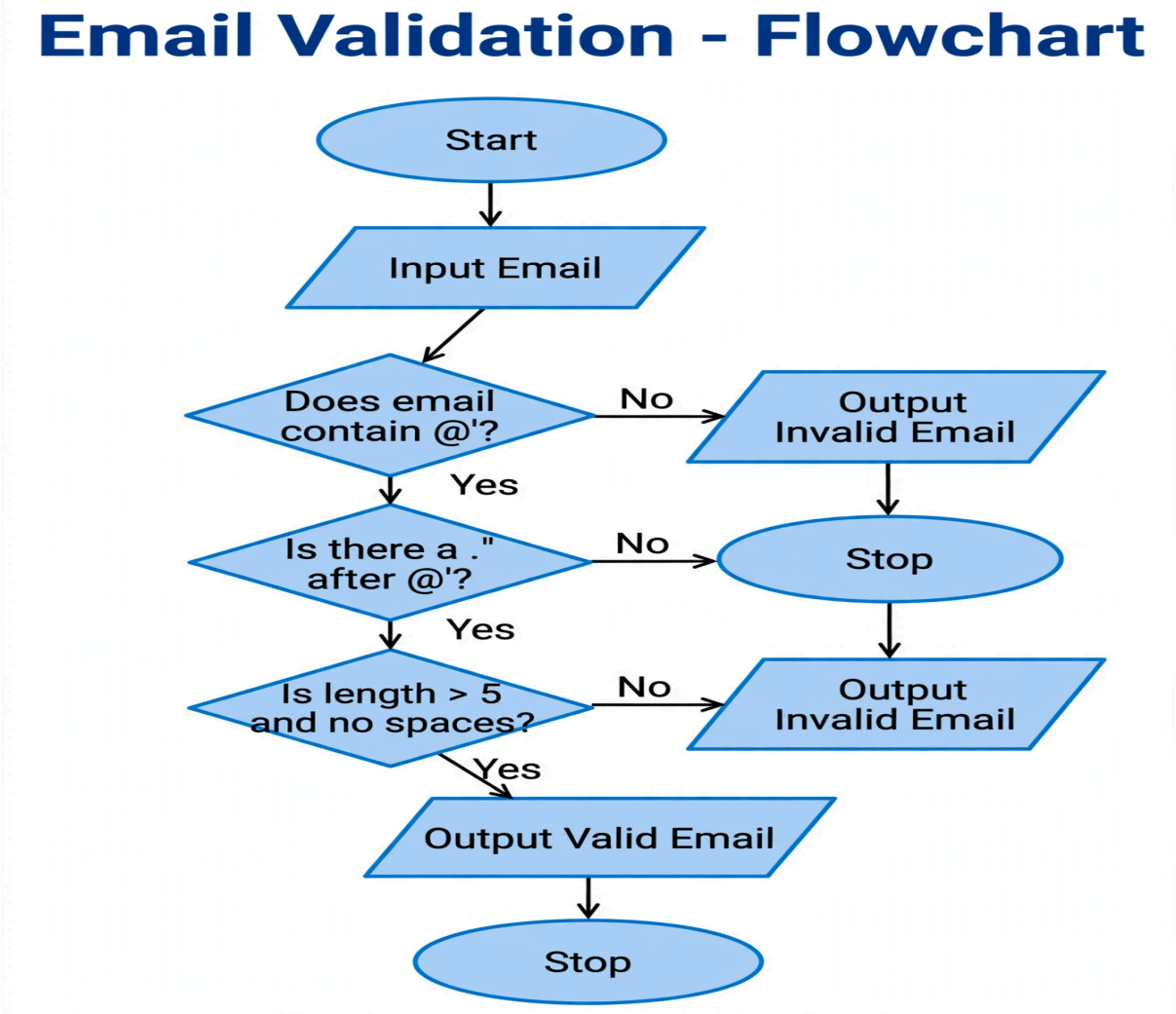
CASE9:Email Validation

Problem Statement: Calculate email validation

Algorithm for email validation:

1.Start the program
2.Ask user for units
3.Check which slab the units fall
4.Calculate bill as per slab rate
5.Show bill amount
6.Stop

Flowchart:



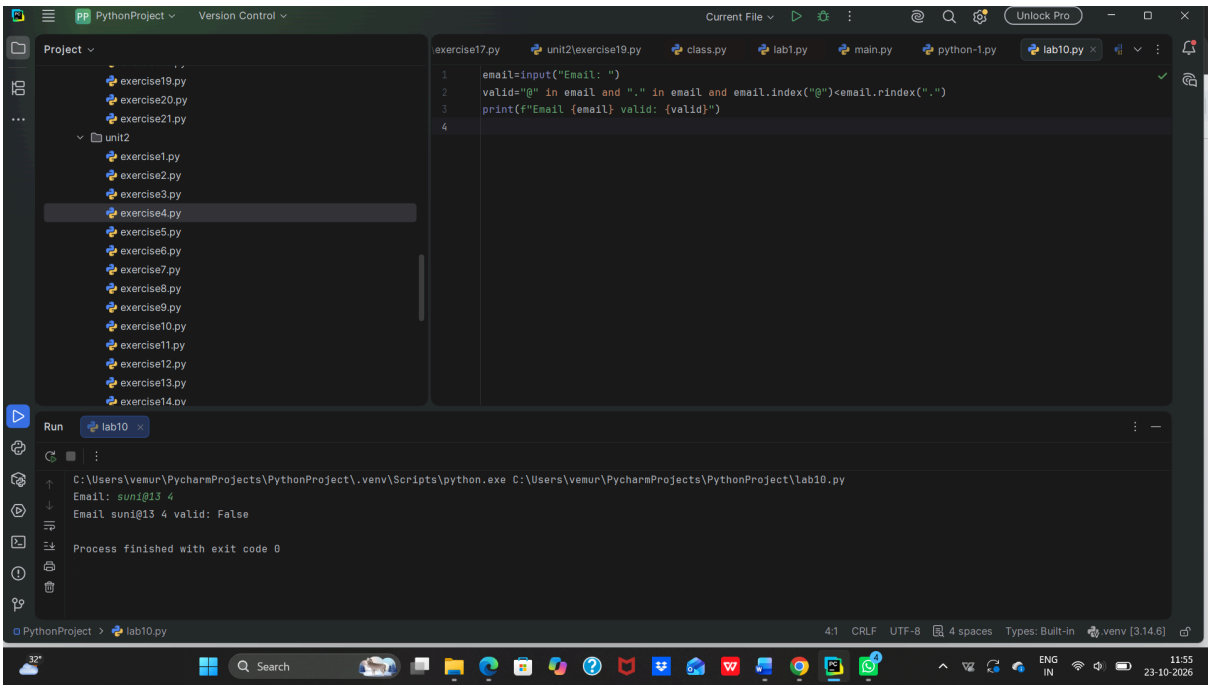
Code:

```
email=input("Email: ")
valid="@" in email and "." in email and email.index("@")<email.rindex(".")
print(f"Email {email} valid: {valid}")
```

Output:
Email: suni@13 4
Email suni@13 4 valid: False

Screenshot:

CASE9:Email Validation



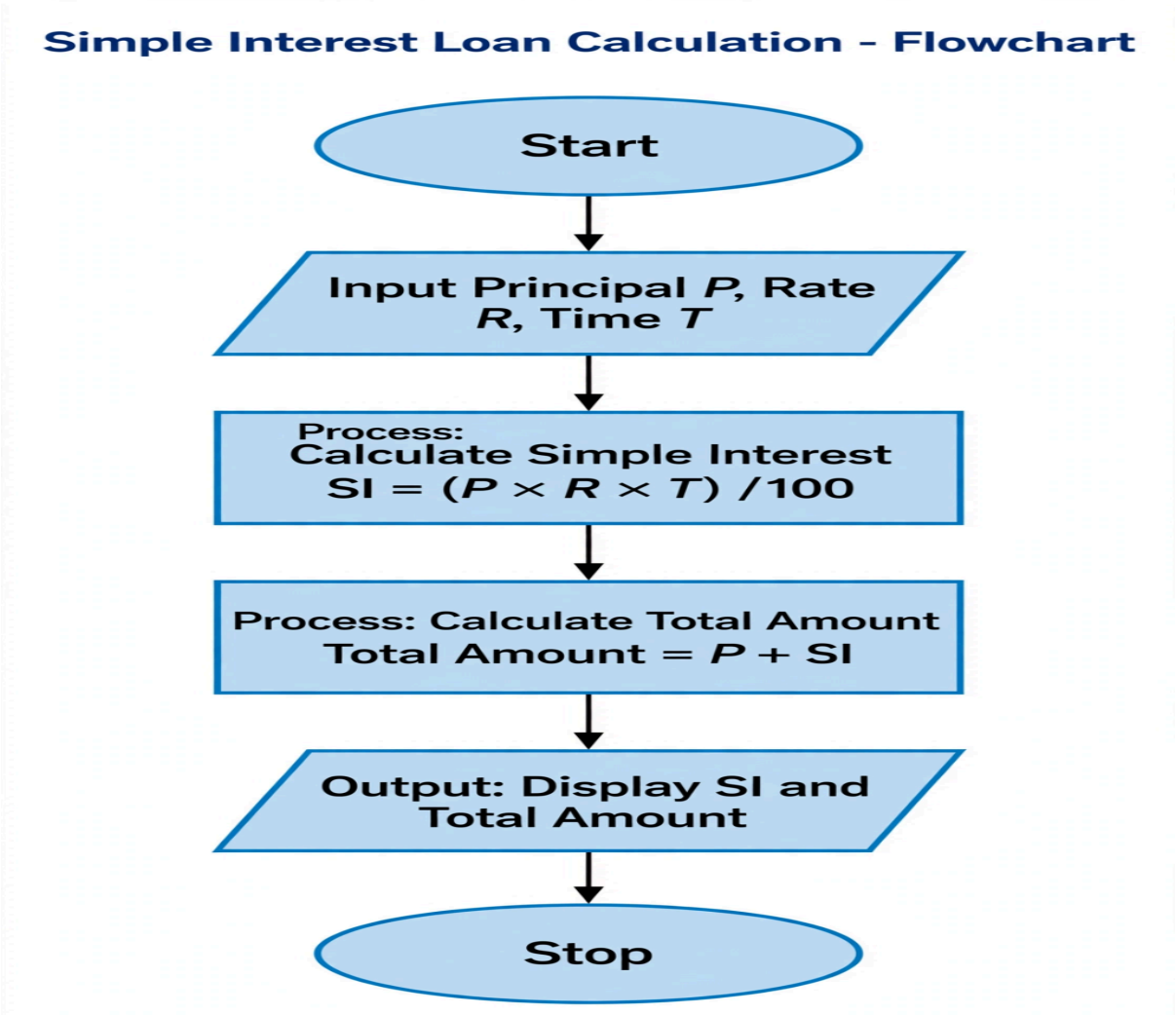
CASE10:Simple Interest Loan

Problem Statement: Calculate Simple Interest Loan

Algorithm for simple interest loan:

1.Start the program
2.Input Principal amount
3.Input Rate of interest
4.Input Time in years
5.calculate simple Interest:si=(p*r*t)/100
6.calculate total amount:amount=p+si
7.Print si and amount
8.Stop

Flowchart:



Code:

```
P=int(input("Principal: "))
R=float(input("Rate: "))
T=int(input("Time: "))
SI=P*R*T/100
amount=P+SI
print(F"SI:{SI} Amount:{amount}")
```

CASE10:Simple Interest Loan

Ouput:
Principal: 1234
Rate: 23.5
Time: 5
SI:1449.95 Amount:2683.95

Screenshot:

